

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) "Coding Requirements" section:

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from a specified volume and load it into Delta tables for customer and order data.

Preconditions:

Source CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

- The Delta tables for customer and order data do not have any existing data or have been truncated.

Main Flow / Functional Steps:

1. Read customer CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` into a Delta table named "customer".
2. Read order CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` into a Delta table named "order".
3. Add a new column "TotalAmount" to the "order" Delta table, which is the product of "PricePerUnit" and "Qty" columns.

FR-DLT-002: Clean and Process Customer and Order Data

Title: Clean and Process Customer and Order Data

Description: The system must remove null and duplicate records from the customer and order Delta tables.

Preconditions:

- Customer and order Delta tables have been loaded with data.

Main Flow / Functional Steps:

1. Remove null records from the "customer" and "order" Delta tables.
2. Remove duplicate records from the "customer" and "order" Delta tables.

FR-DLT-003: Create ordersummary Table and Load Data

Title: Create ordersummary Table and Load Data

Description: The system must create an "ordersummary" table if it does not exist and load data into it by joining customer and order Delta tables.

Preconditions:

- Customer and order Delta tables have been cleaned and processed.
- The "ordersummary" table does not exist or has been truncated.

Main Flow / Functional Steps:

1. Create the "ordersummary" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
2. Join the "customer" and "order" Delta tables on the "CustId" field.
3. Load the joined data into the "ordersummary" table as an SCD Type 2 table.

FR-DLT-004: Update ordersummary Table on Customer Changes

Title: Update ordersummary Table on Customer Changes

Description: The system must update the "ordersummary" table whenever there are changes to the customer data.

Preconditions:

- The "ordersummary" table has been loaded with data.
- Customer data has been updated.

Main Flow / Functional Steps:

1. Identify changes to the customer data.
2. Update the "ordersummary" table by making old records inactive and new records active.
3. Update the "StartDate" and "EndDate" fields accordingly to maintain history.

FR-DLT-005: Create customeraggregatespend Table and Load Data

Title: Create customeraggregatespend Table and Load Data

Description: The system must create a "customeraggregatespend" table if it does not exist and load aggregated data into it from the "ordersummary" table.

Preconditions:

- The "ordersummary" table has been loaded with data.

- The "customeraggregatespend" table does not exist or has been truncated.

Main Flow / Functional Steps:

1. Create the "customeraggregatespend" table if it does not exist in the catalog "gen_ai_poc_databricksco" and schema "sdlc_wizard".
2. Aggregate the "TotalAmount" column from the "ordersummary" table by "Name" and "Date" columns.
3. Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-006: Implement Delta Live Tables**Title: Implement Delta Live Tables**

Description: The system must implement the above steps using Delta Live Tables.

Preconditions:

- All previous steps have been completed successfully.

Main Flow / Functional Steps:

1. Implement the above steps using Delta Live Tables to ensure data is processed and loaded correctly.